

Chemistry

Chemical Formulae of common Compounds :-

Imp

1. Magnesium oxide MgO
2. Sulphuric Acid H_2SO_4
3. Hydrochloric acid HCl
4. Lead Nitrate $Pb(NO_3)_2$
5. Potassium Iodide - KI
6. Zinc sulphate $ZnSO_4$
7. Carbon Monoxide CO
8. Glucose $C_6H_{12}O_6$
9. Barium chloride $BaCl_2$
10. Aluminium sulphate $Al_2(SO_4)_3$
1. Aluminium chloride $AlCl_3$
2. Barium Sulphate $BaSO_4$

13. Sodium Hydroxide $NaOH$
14. Calcium oxide CaO
15. Slaked lime $Ca(OH)_2$
16. Calcium Hydroxide $Ca(OH)_2$
17. Ferrous sulphate $FeSO_4$
18. Ferric oxide Fe_2O_3
19. Calcium Carbonate $CaCO_3$
20. Nitrogen dioxide NO_2
21. Lead oxide PbO
22. Silver chloride $AgCl$
23. Silver Bromine $AgBr$
24. Copper Sulphate $CuSO_4$
25. Copper chloride $CuCl_2$
26. Zinc oxide ZnO
27. Silver Nitrate $AgNO_3$

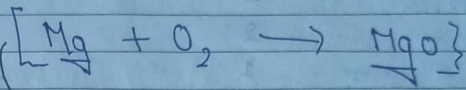
28. Sodium Nitrate NaNO_3
29. Calcium Nitrate $\text{CaNO}_3/2$
30. Aluminium oxide Al_2O_3
31. Ammonia NH_3
32. Aluminium sulphate $\text{Al}_2(\text{SO}_4)_3$
33. Potassium Hydroxide KOH
34. Aluminium oxide Al_2O_3
35. Copper chloride CuCl_2
36. Magnesium Fe_3O_4

Chemical Equations And Reactions

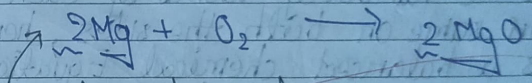
Law of conservation of Mass: Reactants = Products

Reactants $\rightarrow$ Products

1. Magnesium + Oxygen $\rightarrow$ Magnesium oxide



Skeletal / unbalanced chemical equation



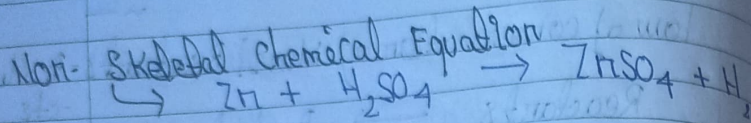
Balanced Equation

Elements	Reactants	Products
Mg	1	1
O	2	1 x 2

Example ①

Zinc + Sulphuric acid $\rightarrow$ Zinc sulphate
+ Hydrogen

Word-equation

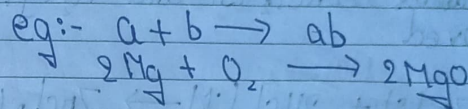


Element	Reactants	Products
Zn	1	1
H	2	2
S	1	1
O	4	4

As the number of atoms of each element is same on both the sides, hence it is a balanced chemical Equation.

Chemical Reaction

Chemical reactions are the process in which new substances with new properties are formed.



(Magnesium + Oxygen) $\rightarrow$ Magnesium Oxide

#

Reactants:-

The substances which take part in chemical reaction are called reactants.

#

Products:-

The new substances produced as the result of chemical reaction are called products.

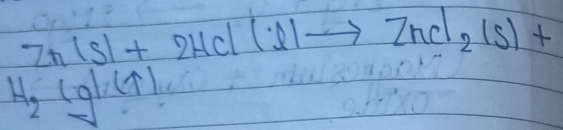
Characteristics of Chemical Reaction

Evolution of gas
 Change in color
 Change in temperature
 Change in state.

Formation of precipitate

1. Evolution of Gas

When Zinc (Zn) reacts with HCl produces $ZnCl_2$ (s) and H_2 (g) is evolved



II. Change in color

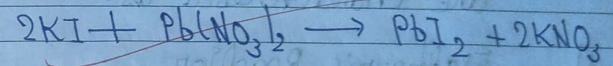
e.g. (a). The chemical reaction between

IV. Change in state

Eg. The combustion reaction of wax due to which solid wax after burning first convert into liquid gas then burns and converts into vapour phase.

V. Formation of precipitate

Eg. Reaction between potassium iodide & lead nitrate produces yellow colored precipitate of lead iodide.

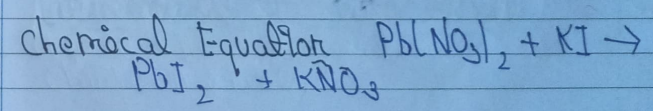


Chemical Equations

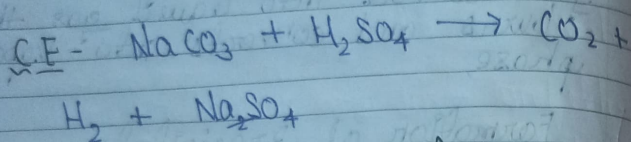
It is a statement which is used to represent a chemical reaction with the help of symbols and formulas

Examples

a). Word Equation - lead nitrate + potassium iodide gives lead iodide + potassium nitrate.



b) W.E - sodium carbonate + Sulphuric acid $\rightarrow$ carbon dioxide + hydrogen + sodium sulphate



Nature of Reaction

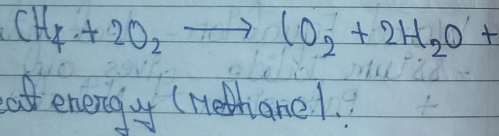
Generally chemical reactions absorb or releases heat due to which they are characterised as:-

- 1) Exothermic Reaction
- 2) Endothermic Reaction

1) Exothermic Reaction - Reactions

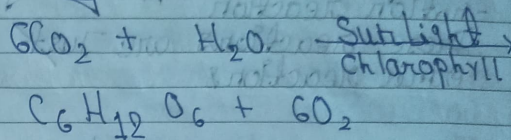
in which heat energy are called Exothermic Reaction

Eg) Burning of natural gas



II) Endothermic Reaction - Reaction in which energy is absorbed for the formation of products is called Endothermic

Eg) Photosynthesis

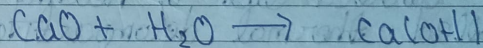


Types of Chemical Reactions

- 1) Combination Reaction $\rightarrow$ when two or more reactions combine to form a single product are called combination reaction

These reactions are exothermic because they releases energy.

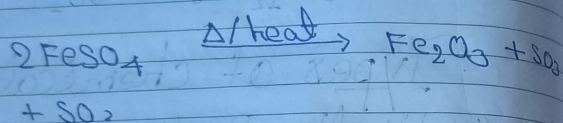
Eg) Calcium oxide + water $\rightarrow$ calcium hydroxide



- 2) Decomposition Reaction

Reactions in which compound splits up into 2 or more simple substances are called decomposition reaction.

- They are just opposite of combination reaction.
- These reactions are just endothermic reactions.

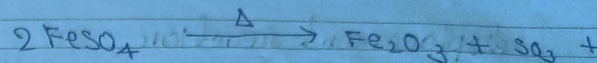
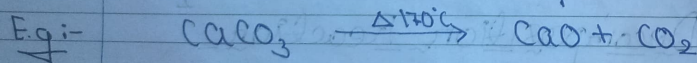


Types of Decomposition (3)

- 1) Thermolysis / Thermal decomposition
- 2) Photolysis
- 3) Electrolysis

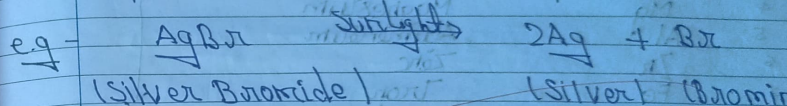
1) Thermal Decomposition

When the decomposition reaction is carried out by heating it is called thermal decomposition.



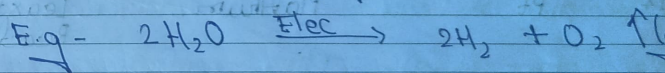
II) Photolysis / Photochemical Decomposition

When chemical substances decompose in the presence of sunlight is called photochemical decomposition.



III) Electrolysis

Decomposition which takes place by passing electricity is called electrolysis / chemical decomposition.



Displacement Reaction

The chemical reactions in which one element takes the place of other element in a compound due to its reactivity is called

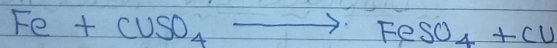
displacement reaction.

Double-Displacement Reaction

Reactivity series of Metal

1.	K	Potassium	Most
2.	Na	Sodium	reactive
3.	Ca	Calcium	
4.	Mg	Magnesium	
5.	Al	Aluminium	
6.	Zn	Zinc	
7.	Fe	Iron	
8.	Pb	Lead	Less
9.	H	Hydrogen	Reactive
10.	Cu	Copper	
11.	Hg	Mercury	
12.	Ag	Silver	
13.	Au	Gold	
14.	Pt	Platinum	Least
			Reactive

Example of Displacement Reaction:

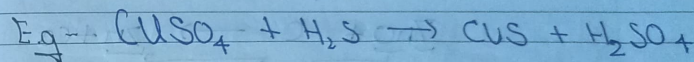


Iron is more reactive than copper
so, iron can displace copper from
copper sulphate.

Double Displacement Reaction

2 compounds react to form 2 other compounds by mutual exchange of ions is called Double displacement reaction.

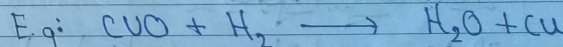
In this reaction precipitate is produced so it is also known as precipitate reaction.



Oxidation

It is the process when:-

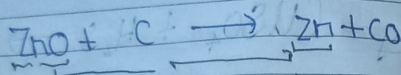
- Oxygen is added to an element or compound.
- Hydrogen is removed to an element or compound.
- Addition of Positive charge
- Removal of Negative charge



Oxidation Ex: $ZnO + C \rightarrow Zn + CO$
Zinc/Carbon is getting oxidised.

Reductions

- When Hydrogen is added to an element or compound.
- When oxygen is removed from an element or compound.



Zinc is getting reduced so it is a reduction.

4. Oxidising Agent

It is a substance which gives oxygen or removes hydrogen and causes the oxidation of substance.

Eg - H_2SO_4 , MnO_2 , etc.

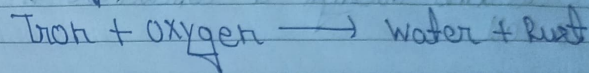
Reducing Agent

It is a substance which hydrogen or remove oxygen and causes reduction.

Corrosion

It is a surface deterioration/degradation process of metals in which they convert to a more stable form due to attack of atmospheric gases.

Rusting



Rancidity

It is the oxidation of oil or fat containing food items when they are kept in the open for a long time resulting in an unpleasant odour or taste.

chips packets filled with nitrogen gas to prevent oxidation of oil or fat containing food items.

by keeping it in airtight containers or by adding anti-oxidants.

ways of preventing - Painting, oil paints, galvanising by iron.

Master's
signature